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Ans: C

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

$$\frac{\pi/3}{2\pi} = \frac{0.050}{\lambda}$$

$$\lambda = 0.30 \text{ m}$$

$$\begin{aligned} v &= f\lambda = 500 \times 0.30 \\ &= 150 \text{ m s}^{-1} \end{aligned}$$

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Ans: A